

Digital LCR Meter
MCR-5000 Series

User manual: <https://wendy4197.github.io/MATRIX/MCR5000-UserManual.pdf>

Programming protocol: <https://wendy4197.github.io/MATRIX/MCR5000-ProgrammingProtocol.pdf>

Software: <https://drive.google.com/file/d/10B3fdbCl6mD-dda8DnTYpg4gKSDug4fi/view?usp=sharing>



- With new 32-bit core processor, the data is as good as the one of foreign first-class equipment
- 4.3 inch true color TFT display
- 30Ω, 100Ω two types of different signal source output resistance
- Built -in comparator, 5 grade flexible quanlified and defective alarming mode shutdown automati cally save test conditions,humanized operation
- Internal more than 100 sets of settingsfiles, U disk extension to save multiple groups of test or recall simplified Chinese, English is optional
- U disk copy screen function, test data saving function, support FAT16 format FAT32 file system, standard with RS232 USB HOST
- USB DEVICE,headphone interface, foot pedal interface and GPIB are optional

Standard

USB Host☒ RS-232☒ Handler(MCR5100/5200)☒

Optional

USB Device☐ GPIB☐ Earphone Jack☐ Foot Pedal☐

Model	MCR-5010	MCR-5030	MCR-5100	MCR-5200
Rated input voltage	AC 220V/110V±10% 50Hz/60Hz			
Test parameter	L,C, R, Z , D, Q , X , ESR , θ(Deg),θ(Rad)			
Test frequency	100Hz、120Hz、1kHz 、10kHz	100Hz、120Hz、1kHz、 10kHz、20kHz、30kHz	40Hz、50Hz、60Hz、75Hz、 100Hz、120Hz、150Hz、 200Hz、250Hz、300Hz、 400Hz、500Hz、600Hz、 750Hz、800Hz、1kHz、1.5kHz 、2kHz、2.5kHz、3kHz、4kHz 、5kHz、6kHz、7.5kHz、10kHz 、12kHz、15kHz、15.7kHz、 16.2kHz、20kHz、25kHz、 30kHz、40kHz、50kHz、60kHz 、66.6kHz、75kHz、100kHz in total 38 frequency points	40Hz、50Hz、60Hz、75Hz、 100Hz、120Hz、150Hz、200Hz、 250Hz、300Hz、400Hz、500Hz、 600Hz、750Hz、800Hz、1kHz、 1.5kHz、2kHz、2.5kHz、3kHz、 4kHz、5kHz、6kHz、7.5kHz、 10kHz、12kHz、15kHz、15.7kHz 、16.2kHz、20kHz、25kHz、 30kHz、40kHz、50kHz、60kHz、 66.6kHz、75kHz、100kHz、 120kHz、150kHz、200kHz in total 41 frequency points
Basic measurement accuracy	0.15%	0.1%		
Test signal level	0.05V, 0.1V、 0.2V, 0.25V,0.3V、 0.3V, 0.4V, 0.5V, 1V			
Equivalent circuit	In series, in parallel			
Mathematical functions	Percentage deviation			
Range way	Automatic, hold, manual selection			
Trigger mode	Internal, manual, external, bus			
Measure speed (≥1kHz)	High speed: The fastest is 30 times/second, middle speed: 10times/second,low speed: 3times/second			
Average time	1—255			
Delay time	0—6s, step is 1ms			
Calibration function	Open circuit/ short circuit / quick reset			
Display mode	Direct reading , Δ% , V/I (Measured voltage/current monitoring)			
Displayer	5 digit display of main and minor parameters , 4.3 inch true color LCD displayer			
Output impedance	30 Ω, 100Ω optional			
Display range				
Z , R , X , ESR	0.1mΩ — 99.999MΩ			
C	0.01 pF — 9.9999F			
L	0.01 μH — 99999 H			
D	0.0001 — 9.9999			
Q	0.0001 — 9999.9			
θ (Deg)	-179.99° — 179.99°			
θ (Rad)	-3.1416 — 3.1416			
Δ%	-999.99% — 999.99%			
Others				
Comparator function	5 grades of sorting function((Except MCR5010))			
Storage	More than 100 sets of internal instrument settings for storage/call, U disk extension of more than 500 sets			
Port	RS232、HANDLER(Except MCR5010/5030)、USB HOST are standard			
Net weight (kg)	3.5			
Gross weight (kg)	4.5			
Instrument size (W*H*D)	240*100*330			
Packing size (W*H*D)	330*210*425			